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Leaf quantity governs understory microclimate, but vertical arrangement takes over where the canopy saturates: a LiDAR-MuSICA attribution

Abstract

Forest understories buffer macroclimatic extremes, yet which dimension of canopy structure drives this buffering — total leaf quantity versus the vertical arrangement of foliage — remains debated. We combine LiDAR-derived canopy traits (LAI, maximum height $H_{\max}$, fractional cover f_{Cover} , and the vertical leaf-area-density profile) with the MuSICA biophysical canopy model to attribute simulated summer understory microclimate ($\Delta T_{\max}$, the diurnal micro-macro temperature offset) at a deciduous oak forest (Blois, France; summer 2021; 53 HOBO loggers). Across **400 LiDAR plots stratified into four structural archetypes** (100 each), we compute an **exact Shapley** attribution of $\Delta T_{\max}$ over all 2^4 trait combinations, toggling each trait against its

own archetype’s realistic canopy. The dominant driver is **density-dependent**: leaf quantity (LAI) governs microclimate variation in **open** canopy ($|\varphi|$ up to 0.41 °C), but where the canopy is **dense** and leaf quantity saturates, the **vertical profile (LAD) becomes the leading driver** ($|\varphi| \approx 0.13$ °C while LAI and cover collapse to ~ 0). A common-baseline landscape view shows leaf quantity and cover dominating with the vertical profile ≈ 0 *on average* — but that average **hides the dense-canopy regime** that the realistic per-archetype baseline reveals; the vertical profile is a small absolute lever everywhere, leading in dense canopy only because quantity saturates. MuSICA reproduces the between-plot buffering ranking well ($r \approx 0.93$) but hits a glass ceiling on absolute level (+0.9 °C warm bias) and on topographic / sub-pixel canopy-gap heterogeneity a 1-D plot model cannot represent. This density-dependent control connects to the canopy-density threshold of neighbouring work and motivates the optical-saturation analysis of Chapter 2.

1. Introduction

- Forest microclimate is decoupled from the regional macroclimate and is the climate organisms actually experience: understory conditions can differ from open-field / regional weather by several degrees, with direct consequences for forest biota [REF De Frenne et al.; Lembrechts et al.; Zellweger et al.].
- This buffering shapes forest ecosystem functioning — it underpins thermal refugia, tree regeneration, understory biodiversity and species persistence under climate warming [REF].
- Canopy structure is the primary driver of buffering [REF]. Traditionally characterised from field inventories, it can now be derived from remote sensing — in particular airborne laser scanning (ALS/LiDAR), which resolves the full 3-D vegetation profile — and ALS-derived structure has been used in forest-microclimate studies [REF Zellweger; Jucker]. In a neighbouring LiDAR + MuSICA study, Bouwen [REF] reported a canopy-density threshold on buffering, leaving open whether the same holds in deciduous stands and, more fundamentally, **which structural dimension actually drives the effect**.
- But, more precisely, which structural variables contribute most to buffering remains unclear. LiDAR is valued *precisely* for resolving the vertical profile, and the vertical arrangement of foliage is widely assumed decisive — **yet whether the vertical dimension itself matters, or merely co-varies with the total quantity of leaves (LAI) and cover, has never been disentangled**. This is our core question: leaf quantity vs vertical arrangement.
- Statistical modelling of microclimate (linear offset / sensitivity “slope”, equilibrium models, GAMMs) is data-cheap, predictive and widely used [REF] — but it cannot cleanly *attribute* buffering to individual structural traits: canopy traits are strongly collinear, so regression coefficients are unstable (concurvity) and the effect of any one trait cannot be isolated observationally.
- A physics-based proposition addresses exactly this: a mechanistic multilayer canopy model (MuSICA [REF]) solving the energy / radiative / turbulent balance can simulate **controlled** trait combinations, attributing buffering to physical traits and disentangling correlated variables that a regression cannot separate — and being process-based, it is transferable in principle.
- In this study, we couple ALS-derived canopy traits with MuSICA and apply an exact-Shapley attribution over all trait combinations, validated against 53 HOBO loggers at Blois, to quantify the relative contribution of leaf quantity vs vertical arrangement to summer understory buffering.

We hypothesised that: - **(H1)** vertical structure modulates radiative / turbulent transfers, so the LAD profile is expected to be a key parameter for faithful buffering simulation; - **(H2)** at fixed LAI and height, top-heavy profiles (biomass concentrated high) cause stronger diurnal attenuation (radiation intercepted higher, heat sources shifted to the canopy top).

To achieve our objective and test these hypotheses, we: (1) derived a structural typology of the LAD profile (FPCA + k-means $\rightarrow$ archetypes P1-P4); (2) simulated the 2^4 trait coalitions per plot

with MuSICA and computed an exact-Shapley attribution of $\Delta T_{\max}$, with an off-manifold conditional control for correlated traits; and (3) validated the simulated buffering against the HOBO network and decomposed the residual (the glass ceiling) against topographic / heterogeneity covariates that MuSICA cannot see.

As a perspective toward Chapter 2, we further ask whether a 2-D optical proxy (Sentinel-2) can substitute for LiDAR structure or merely gives an optical *illusion* of buffering (a structure proxy).

2. Material and Methods

2.1 Study site and in-situ microclimate

- Blois is a temperate lowland deciduous forest located in the centre of France, almost entirely composed of sessile oaks (*Quercus petraea*).
- 53 HOBO loggers, distributed along a canopy light / structure gradient, record hourly air-temperature measurements at 1 m above the ground (summer 2021).
- The macroclimate reference is open-field / regional weather; $\Delta T_{\max}$ = diurnal micro – macro temperature offset (the buffering metric).

2.2 LiDAR canopy structure and traits

- 0.5 m vertical leaf-area-density (LAD) profiles from ALS (resolving the understory); traits: LAI, maximum height Hmax, fractional cover fCover, VCI, and profile shape.
- LiDAR LAI is one-sided (projected) → **×2 = two-sided leaf area** expected by MuSICA (not a clumping correction).

2.3 Structural typology

- FPCA of the **doubly-normalised LAD profile** (normalised in height x AND density y → pure shape, scale-free of LAI / Hmax); K-means on the FPCA scores → **four archetypes P1-P4** (architectural-shape-based, independent of leaf amount). Clustering frozen; cLHS design.

2.4 MuSICA model and simulation design

- MuSICA is a multilayer biophysical canopy model; it ingests **LAI as two-sided leaf area distributed per layer** and the **LAD profile as the vertical discretisation** of that leaf area.
- **Design = per-point cLHS, stratified by archetype**: a conditioned Latin-hypercube sample of **400 plots (100 per archetype P1-P4)** spanning the real trait space; **16 coalitions (2⁴) per plot** (each trait real vs baseline). 6 400 MuSICA runs; legacy binary **v3.2.0**.
- **Two baseline schemes (the choice IS the question)**:
 - *Main — cluster-specific baseline*: each trait toggled against **its own archetype's mean** (LAD→uniform). Realistic / on-manifold per type; coalition 0000 = that archetype's canopy. → within-archetype drivers.
 - *Complement — common (global) baseline*: LAI→mean (LAI_b ≈ 6.73), Hmax→mean, fCover→mean (0.869), LAD→uniform. → landscape attribution (but off-manifold for atypical types — see R2).
- *Why per-point, not the 4 mean profiles*: MuSICA is nonlinear, so the Shapley of a cluster's **mean profile** is Jensen-biased ($f(\text{mean}) \neq \text{mean}(f)$); per-point + aggregation is unbiased and yields the within-archetype distribution (the Jensen gap is itself reported: largest in the sparse P1).

2.5 Microclimate metric extraction (two conventions)

- (a) absolute $\Delta T_{\max}$ / $\Delta VPD_{\max}$ $\rightarrow$ fixed 1 m interpolation, -2 h shift (HOBO-comparable);
- (b) micro-macro slope (buffering / amplification split) $\rightarrow$ $n_{\text{air}}=1$, no shift.

2.6 Shapley attribution (all combinations of traits)

- **Exact Shapley** ϕ per trait per plot ($\Delta T_{\max}$, $\Delta VPD_{\max}$), additivity verified ($\Sigma \phi = \text{full} - \text{base}$, to machine precision), bootstrap CIs; isolates ϕ_{LAD} vs ϕ_{LAI} in place of the GAMM.
- **Main readout (cluster-specific baseline):** mean $\phi \approx 0$ by design, so importance = mean $|\phi|$ per archetype $\rightarrow$ within-archetype trait importance.
- **Complement (common baseline):** signed $\phi \rightarrow$ landscape ranking (report median / $|\phi|$, **not** the design-pooled signed mean, which the broad cLHS tail skews).
- **Conditional Shapley** bounds the off-manifold artefact of correlated traits on the common-baseline run: $\phi_{\text{LAD}} \mid H_{\max}$ (structural co-definition) and $\phi_{\text{LAI}} \mid f_{\text{Cover}} \ \& \ \phi_{f_{\text{Cover}}} \mid \text{LAI}$ (statistical collinearity $r = 0.76$) — gaps ≤ 0.02 °C.

2.7 Validation and spatial-heterogeneity decomposition

- REF vs 53 HOBO; daily $\Delta T_{\max}$; micro-macro slope (45/8 split); Moran's I for spatial autocorrelation.
- $\Delta VPD_{\max}$ is attributed but not validated (HOBO measure air temperature only); VPD results are model-internal.
- The validation residual (HOBO – model $\Delta T_{\max}$) is regressed on topographic / heterogeneity covariates from an external LiDAR dataset (elevation/DTM, slope, TWI, northness, canopy height-SD, rumple, gap fraction) — variables a 1-D plot model cannot represent.

3. Results

R1 — A four-archetype structural typology

Four archetypes from open **P1** to dense **P4** (**Fig 1**; + `tab_cluster_structure`). Although the typology is built on profile *shape* (FPCA of the doubly-normalised LAD), the four archetypes line up along a **canopy-density / LAI gradient** (P1 sparsest $\rightarrow$ P4 densest). The typology is therefore a presentation device that organises this gradient and lets us show *how the attribution redistributes along it* (R2, Fig 3) — it is **not** the engine of the attribution, which is computed per plot, independently of the clustering.

R2 — Which trait governs understory microclimate is density-dependent (*core*)

Main analysis — within-archetype drivers (per-point cLHS, cluster-specific baseline; Fig 2). For each archetype we run an exact Shapley on **100 cLHS plots**, toggling each trait against **its own cluster's realistic mean** (not a global mean, which would impose off-manifold canopies on atypical types — e.g. a landscape-mean dense canopy on a sparse-tree cluster). Mean $\phi \approx 0$ by design, so importance = mean $|\phi|$. The leading driver of plot-to-plot $\Delta T_{\max}$ variation **shifts with canopy density**: - **Open canopy (P1):** LAI dominates ($|\phi| = 0.41$ °C), with f_{Cover} (0.21), LAD (0.17) and $H_{\max}$ (0.13) all contributing — the most variable type. - **Intermediate (P2, P3):** LAI leads (0.20, 0.15); others minor. - **Dense canopy (P4):** LAI and f_{Cover} **saturate** ($|\phi| = 0.10$ and 0.01) $\rightarrow$ the **vertical profile (LAD) becomes the leading driver** ($|\phi| = 0.13$ °C). Where leaf quantity can no longer vary the microclimate, *arrangement* governs the residual. $\rightarrow$ **Leaf quantity drives the open end of the gradient; the vertical profile governs the dense end**

where quantity saturates — a density-dependent control that connects directly to Bouwen’s density threshold.

Complement — landscape attribution against a common baseline (Fig 3, Annex). With a *single* baseline across the whole landscape, leaf quantity and cover dominate and $\phi_LAD \approx 0$ on average ($\phi_LAI \approx \phi_fCover \approx -0.2$ °C; $\Sigma\phi \approx -0.38$; same order for VPD). This is what a single-baseline Shapley (or the V1 GAMM) reports, and the conditional/off-manifold control (A4/A4b) confirms the collinearity ($LAI \leftrightarrow fCover$ $r = 0.76$) does not distort this global ranking (gaps ≤ 0.02 °C). But this common baseline is **off-manifold for atypical types** and averages the density structure away: only the realistic per-archetype baseline reveals that **“LAD ≈ 0 ” is a landscape average, not a universal — it hides the dense-canopy regime where the profile leads.**

Reconciling with H2 (no contradiction). The vertical profile is a **small absolute lever everywhere** ($|\phi_LAD| \leq 0.17$ °C; H2’s top-heavy mechanism is real but second-order and saturates — Annex A2/A3, field partial $r = -0.04$ at full control). It “dominates” in dense canopy only **relatively**: LAI and fCover collapse to ~ 0 leverage there, so the small vertical effect is what remains. **Small-but-leading, not large.**

R3 — The 3-D distributes the variance, but a glass ceiling against the field

Between-plot ranking captured ($r \approx 0.93$; Fig 4); warm **bias +0.9 °C**; field-HOBO split 45/8, z05-model split 46/7. Per-plot slope, MuSICA vs HOBO, splits buffering/amplifying: the model compresses the slope range toward 1 — under-buffers the bufferers (sim 0.94 vs obs 0.84) and under-amplifies the amplifiers (1.12 vs 1.27). The glass ceiling is two things: (i) real structural under-buffering on the closed-canopy plots (gap ≈ 0 , topo negligible) — the genuine model limit — and (ii) an out-of-scope inability to reproduce topo/gap-driven amplification on the ~ 8 amplifying plots.

Decomposing the residual (what the model misses; Fig 5, Annex A8). Regressing the per-plot validation residual (HOBO – model ΔT_{max}) on topographic / heterogeneity covariates explains **$R^2 = 0.51$ ($p = 3e-4$)** — about half the glass ceiling is structured, not noise. The dominant term is **canopy gap fraction** ($r = 0.58$, $p < 0.001$), with secondary topographic exposure (elevation/DTM $r = -0.29$, slope +0.31, northness -0.31 ; all $p \approx 0.04$). Crucially, **vertical canopy heterogeneity does not contribute** — CHM height-SD ($r = 0.04$, ns) and rumple ($r = 0.17$, ns) are null. The 8 amplifying plots are the gappy/steeper/more exposed ones (gap 0.050 vs 0.002; slope 4.9 vs 1.7). So the unreproduced amplification is **sub-pixel canopy openness + topographic exposure** — horizontal, not vertical. The attribution and buffering hierarchy also hold for night-time ΔT_{min} (one sentence; daytime ΔT_{max} is the focus).

4. Discussion

D1 — A density-dependent control: quantity in the open, arrangement in the dense

The leading driver shifts with canopy density (R2). In open canopy, near-unsaturated interception means **total leaf quantity** sets the microclimate — adding leaves still changes radiative/turbulent transfer, so LAI dominates. As the canopy closes, interception/extinction **saturate**: extra leaf area and cover no longer move ΔT_{max} ($|\phi_LAI|$, $|\phi_fCover| \rightarrow \sim 0$ in P4), so the remaining variation is governed by **where** the foliage sits — the **vertical profile**. This is exactly Bouwen’s density threshold expressed at the trait level: below threshold, quantity; above it, arrangement. Crucially the vertical effect is **small in absolute terms everywhere** ($|\phi_LAD| \leq 0.17$ °C; Canopy-Ratio adds nothing beyond LAI) — it *leads* in dense canopy by saturation of the others, not by being large. The vertical gradient of wind / RH / VPD (Fig 6, mechanistic illustration — no vertical field reference) shows the transfer mechanism operating but saturating. H2 is the clean illustration of this: its mechanism is real (top-heavy cools more, in silico) yet operationally negligible — the COM variation available at fixed structure is narrow (residual SD ≈ 0.09

vs the 0.50 span imposed in silico), so the expected field effect ($\approx 0.05\text{--}0.10\text{ }^{\circ}\text{C}$) is $\sim 10\times$ below the between-plot noise ($\text{SD} \approx 0.65\text{ }^{\circ}\text{C}$) and undetectable. The naive field signal (COM partial $r = -0.43$ controlling LAI only) was COM proxying fCover ($r = 0.62$) and Hmax ($r = 0.47$); it vanishes at full control (-0.04). No model $\leftrightarrow$ reality disagreement: the profile acts in the right direction but is not an operative lever in real stands.

D2 — The glass ceiling and what a 1-D model cannot see

MuSICA captures the between-plot ranking ($r \approx 0.93$) but not the absolute level: the $+0.9$ to $+2\text{ }^{\circ}\text{C}$ warm bias is most likely a **forcing artefact** (ERA5 above-canopy reanalysis input), not a process error in the canopy scheme (secondary candidates: soil moisture, sensor radiative error); to be revisited if the forcing is corrected. MuSICA is a plot-scale **1-D** model — slope, aspect, elevation, cold-air pooling, lateral advection, edge effects and soil-moisture heterogeneity are not inputs. These do **not** bias the attribution (absent from the 16 coalitions — φ stays valid within the MuSICA world); they are confined to the validation residual, which is exactly why the residual decomposition (R3: $R^2 = 0.51$, gap-fraction-dominated, vertical heterogeneity null) localises the missing amplification in **horizontal openness + topographic exposure**. This is the spatial-heterogeneity mirror of $\varphi_{\text{LAD}} \approx 0$: the vertical dimension is inoperative both inside the model (attribution) and outside it (residual). Note fCover (the plot-mean cover MuSICA ingests and the Shapley attributes) and gap fraction (sub-pixel heterogeneity a single plot-mean cannot represent) are different variables, so gap fraction explaining the residual is not the attribution leaking back in. Spatial structure is genuine but does not inflate the inference (Moran's I): the observed buffering slope is autocorrelated ($I = 0.30$, $p < 0.001$), the model slope reproduces this but dampened ($I = 0.13$, $p = 0.02$ — another facet of the ceiling), and crucially the **residual is not autocorrelated** ($I = 0.09$, $p = 0.07$), so per-plot inference is not pseudoreplicated; cLHS sampling along the structural gradient further mitigates.

D3 — Methodological contribution

Exact Shapley supersedes the GAMM (resolves concurvity, replaces the black-box “Date” with physical ERA5 forcings); it averages over all coalitions and is robust under collinearity, and conditional Shapley bounds the off-manifold artefact on both correlated pairs — $\varphi_{\text{LAD}} \approx 0$ holds on the coherent Hmax-real subset ($+0.022$ vs $+0.023$ off-manifold), and $\varphi_{\text{LAI}} / \varphi_{\text{fCover}}$ are stable whether the collinear partner is real or baseline (gaps $0.022 / 0.010\text{ }^{\circ}\text{C}$), so neither result is a chimeric-coalition artefact. The $-0.43 \rightarrow -0.04$ field contrast demonstrates why full-control Shapley beats naive partial correlations / GAMM partial effects. The deeper rationale for a mechanistic model is physical (energy/radiative/turbulent balance, not a fitted slope or equilibrium model) and generalisable in principle (transferable parameters): statistical microclimate models describe; MuSICA lets us **attribute** to physical traits and, downstream (Ch. 3), **spatialise** dynamically. The attribution is model-internal but spatially validated.

D4 — Limitations and perspectives

Single-site, single-year, single-species scope. Trait ranking is identical for v3.2.0 and v3.2.3 (v3.2.0 keeps the spatial structure, $r\ 0.93$ vs 0.63 ; v3.2.3 only trims the warm bias), so the conclusions are robust to the model version. Toward Chapter 2: the Sentinel-2 optical signal saturates and gives an optical illusion of buffering, and the naive all-satellite run (S2 LAI + FORMS-H height $\rightarrow$ MuSICA) is a Chapter-3 baseline.

5. Conclusion

Which canopy trait governs understory microclimate is **density-dependent**: leaf quantity (LAI) drives the variation in open canopy, but where the canopy is dense and quantity saturates, the **vertical profile takes over** as the leading driver. The vertical dimension is a small absolute lever everywhere — its prominence in dense stands is a saturation effect, not a large effect — which both nuances the “3-D is indispensable” dogma and explains why a landscape average reads “vertical ≈ 0 ”. MuSICA reproduces the between-plot buffering ranking well ($r \approx 0.93$) but plateaus on absolute level and amplifiers → an honest glass ceiling. The density-dependent control connects to the canopy-density threshold of neighbouring work and opens Chapter 2.

Figures

- **Fig 1** typology — LAD profiles of P1-P4 (fig_archetypes_profiles, LAI one-sided)
- **Fig 2** (*core*) within-archetype drivers — $|\phi|$ per trait per archetype, cluster-specific baseline (FigSh_shapley_clhs_cluster); LAI leads open P1, LAD leads dense P4 → density-dependent
- **Fig 3** landscape attribution (common baseline) — leaf quantity & cover dominate, LAD ≈ 0 on average (FigSh_shapley_clhs_global / z05 ranking); the average that Fig 2 refines
- **Fig 4** validation / glass ceiling — REF vs HOB0, $r \approx 0.93$
- **Fig 5** residual decomposition — residual vs gap fraction / topo ($R^2 = 0.51$)
- **Fig 6** vertical gradient — mechanistic illustration (cited in D1)

(forward-validation-in-Shapley-order figure → Annex; two validation figures in the main text would over-weight validation for an attribution paper.)

Annexes

- **A1** FPCA of doubly-normalised LAD shape (→ clustering) + k-means elbow + cluster map (A1a-d)
- **A2** H2 field validation (COM vs buffering; null at full control)
- **A3** H2 controlled in-silico (top-heavy cools more; saturates)
- **A4** conditional Shapley for LAD | Hmax (off-manifold bound, structural co-definition)
- **A4b** conditional Shapley for LAI | fCover & fCover | LAI (off-manifold bound, collinearity $r = 0.76$; gaps ≤ 0.02 °C)
- **A5** extra metrics (ΔT_{min} / amplitude / stability)
- **A6** version robustness v3.2.0 vs v3.2.3 (a scatter, b heatmap, c sim-vs-obs)
- **A7** Sentinel-2 / optical illusion = saturation (Chapter-2 bridge)
- **A8** glass ceiling — slope buffering/amplifying panel (the topography $R^2 = 0.51$ panel is promoted to main Fig 5)
- **A9** sampling-height robustness (nair==1 vs fixed 1 m)
- **A10** trait collinearity matrix (LAI/Hmax/fCover/COM/VCI)
- **A11** forward validation in Shapley order (demoted from main text)